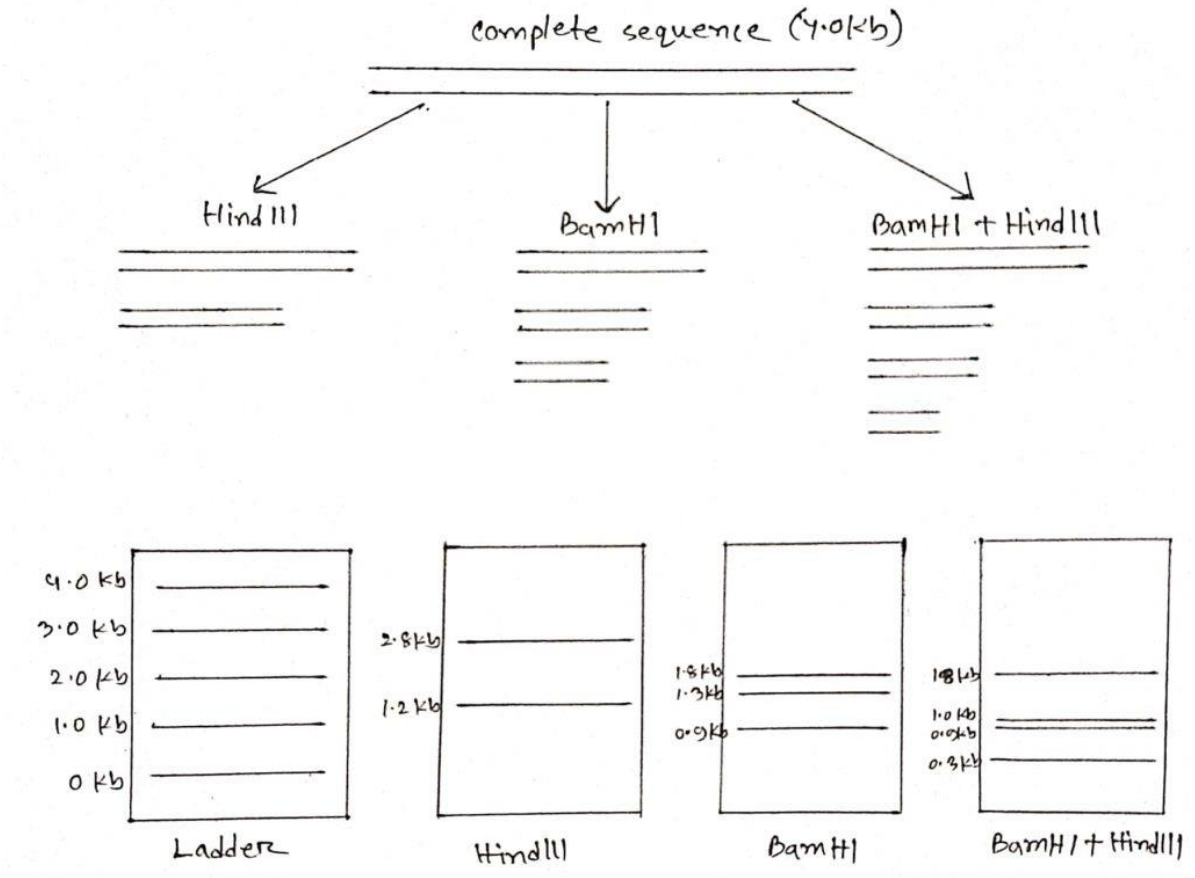
**What is Restriction mapping?**

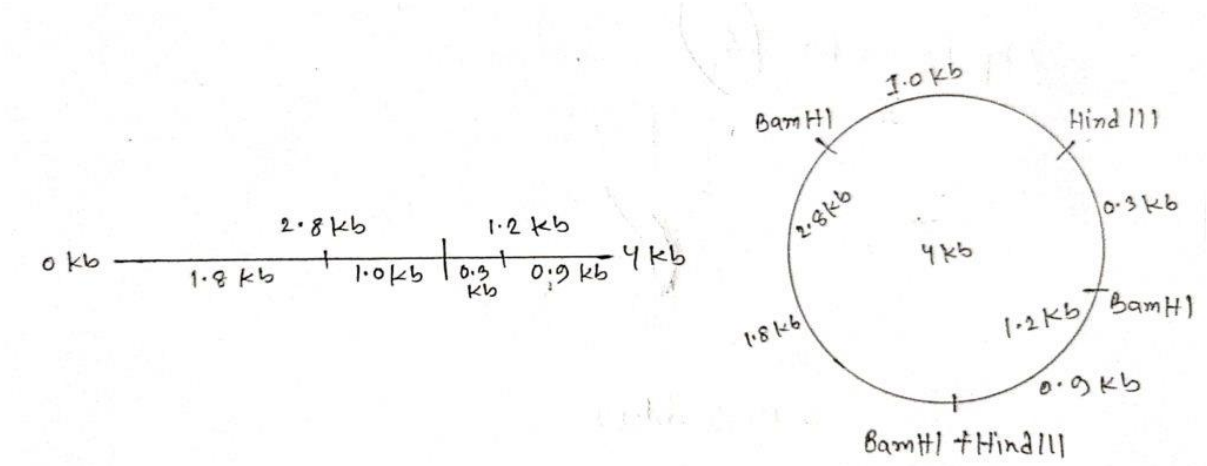
Restriction mapping is a process used to map an unknown segment of DNA by cutting the DNA into pieces with specific restriction enzymes and then identifying the locations of the breakpoints. The enzymes are generally the **endonucleases** that cut the DNA at specific sites.

**Restriction mapping steps:**

- For performing restriction mapping first of all we need a sample DNA sequence. The size of the sample DNA sequence should be known. Here, as an example a size of 4.0 kb complete DNA sequence has been taken.
- The complete DNA sequence is then digested with Hindlll and BamHI. Firstly, single restriction digestion is performed with **Hindlll** and **BamHI** separately. Cutting with Hindlll gives two fragments and cutting with BamHI creates three fragments.
- After that, double restriction digestion is performed with combined Hindlll+BamHI. This creates four fragments of different sizes.
- Then the fragments are run in gel electrophoresis which gives different bands according to the size of the fragments.
- After band visualization and analysis, we can see that the two fragments given by Hindlll are of 2.8kb and 1.2kb in size whereas the three fragments given by BamHI are 1.8kb, 1.3kb and 0.9kb in size. If we analyze the fragments of double restriction digestion we can see that the fragments are 1.8kb, 1.0kb, 0.9kb and 0.3kb.




From the process, we can understand that 1.8kb and 1.0kb fragments in the double digestion together represents 2.8kb fragment and 0.9kb and 0.3kb represents 1.2kb fragment produced by Hindlll in single restriction digestion. Again, 1.0kb and 0.3kb together represents 1.3kb fragment produced by BamHI in single digestion. So, we can identify the relative positions of restriction sites involving in single and double digestion based on the sizes of resultant DNA fragments.




**Applications:**

- Used in the identification of restriction sites
- Used in the study of mutation
- Insert analysis during cloning
- Study of insertion and deletion of a gene

**Limitations:**

- If there are few cutting sites for the enzyme that is being used, it becomes easier to generate restriction maps. However, if so much restriction sites are present, the result sometimes become ambiguous.
- Restriction mapping process is more suitable for smaller DNA molecules rather than larger molecules. If the DNA molecule that is being used is less than 50kb in length then it is possible to construct the map.

**Restriction Endonuclease?**

Restriction endonuclease or restriction enzyme is a protein that cuts DNA at specific sites. They are produced from bacteria. EcoRI, BamHI, HindIII are the examples of restriction endonuclease enzymes.

**Insertion and Deletion**

Insertion is defined as a type of mutation in which one or more nucleotide is added into the DNA sequence. On the other hand, deletion is also a type of mutation which refers to the removal of one or more nucleotides from the DNA sequence.

**Double digestion of DNA**

Double digestion of DNA means digesting DNA fragments with two restriction endonucleases simultaneously. Double digestion helps in the identification of correct orientation of the gene of interest.

**Gel electrophoresis**

Gel electrophoresis is a technique used in the separation and analysis of DNA, RNA, protein etc. based on their sizes and charge.

**Restriction site for BamHI and HindIII?**

GGATCC is the restriction site for BamHI and AAGCTT is the restriction site for HindIII